

Naive RAG vs KAG vs Hybrid

Twenty-nine questions put to three retrieval architectures over the same three-page PDF. Every answer below is the one the system actually gave – including the wrong ones, which are the point.

Provenance. Naive RAG and Hybrid answers are verbatim LLM outputs recorded from live runs, stored in `data/naive_rag_results.json` and `data/hybrid_results.json`. KAG answers are deterministic Cypher results – the queries live in `src/kag.py`, the facts they traverse in `src/graph_data.py`. Nothing here is reconstructed from memory.

Top-*k* retrieval is a **sample** of the corpus, so it cannot answer questions about the whole corpus. An ontology is a **projection** of the text, so it cannot answer questions about what it did not model.

THE RESULT

Neither pure approach wins

They fail in opposite directions, and the failures are structural rather than incidental – which is why adding a better embedding model does not fix the middle row, and a richer ontology does not fix the bottom one.

Measured on 18 questions, one document, the same LLM throughout.

QUESTION TYPE	NAIVE RAG	KAG	HYBRID
Local facts — negation, OR/AND logic, absence	CORRECT	CORRECT	CORRECT
Global aggregates — counts, exhaustive lists	WRONG	CORRECT	CORRECT

Unmodelled prose — rationale, objectives, description

CORRECT

NO DATA

CORRECT

Context recall — does the evidence even reach the prompt? Measured with no API calls.

CONTEXT ASSEMBLED FROM	LOCAL	GLOBAL	PROSE	TOTAL	PROMPT SIZE
passages only	2/2	11/21	4/4	63%	2,337 chars
graph only	2/2	21/21	0/4	85%	3,611 chars
hybrid	2/2	21/21	4/4	100%	5,951 chars

Hybrid reaches 100% at **2.5× the prompt size**. That is the trade, and it is not free.

Naive RAG — passages only

KAG — graph only

Hybrid — graph + passages

✓ CORRECT

✗ WRONG

⚡ NUANCED

Ø NO DATA

SET 1 · 18 QUESTIONS

Naive RAG against the graph

The baseline comparison. Question ids are the ones used in the repository, so each block can be traced straight back to `data/naive_rag_results.json`. C1b, C2b and C3b are paraphrase variants written to probe the RAG baseline and have no Cypher counterpart.

LOCAL FACTS — BOTH APPROACHES CORRECT

c1 Which entrance exam must a B.Tech Fashion Technology applicant take?

IN THE DOCUMENT None — the document states there is *no* entrance examination.

NAIVE RAG ✓ CORRECT

“There is no entrance examination for the B.Tech Fashion Technology Programme.”

KAG · GRAPH ✓ CORRECT

required_exams: 0, plus an explicit EXEMPT_FROM → VITEEE edge carrying the evidence “There is NO Entrance Examination.”

WHY The absence is stored as an *edge*, not as a missing edge — so it can be told apart from M.Des, where the route is simply unstated.

c1b List every B.Tech programme and state the entrance exam each one requires.

IN THE DOCUMENT CSE / EEE / ECE / Civil / Mechanical → VITEEE; Fashion Technology → none.

NAIVE RAG ✓ CORRECT

“For B.Tech Fashion Technology, there is no entrance examination. For B.Tech CSE, EEE, ECE, Civil, and Mechanical programs, students need to undertake the VIT Engineering Entrance Exam (VITEEE).”

KAG • GRAPH – NOT RUN

No Cypher written — C1b is a paraphrase variant used only to probe the RAG baseline.

c2 Do I need both a UCEED score and a V-DAT score to get into B.Des?

IN THE DOCUMENT No — UCEED *or* V-DAT, plus 10+2 marks.

NAIVE RAG ✓ CORRECT

“No, you do not need both. Selection for B.Des is based on a valid UCEED score *or* a V-DAT score, along with 10+2 marks in the qualifying examination.”

KAG • GRAPH ✓ CORRECT

OR_1 → [UCEED score, V-DAT score], read as **any one of these**; AND → [10+2 marks].

WHY The `condition_group` property is what makes OR/AND machine-readable instead of a matter of prose interpretation.

c2b I have a UCEED score but no V-DAT score. Am I eligible for B.Des?

IN THE DOCUMENT Yes — they are alternatives.

NAIVE RAG ✓ CORRECT

“Yes, you are eligible. Selection is based on a valid UCEED score *or* V-DAT score, with 10+2 marks.”

KAG • GRAPH – NOT RUN

Paraphrase variant — RAG baseline only.

c3 How is a student admitted to M.Des?

IN THE DOCUMENT Not stated — the source is truncated mid-sentence on page 3.

NAIVE RAG ✓ CORRECT “The context provides admission information for the B.Des program but does not contain details on how a student is admitted to the M.Des program.”	KAG • GRAPH ✓ CORRECT status: NOT_STATED_IN_SOURCE , reason: “source document truncated mid-sentence on page 3” WHY The graph records <i>why</i> the fact is absent, so the query returns a reason rather than a bare empty result.
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c3b What score does M.Des require?

IN THE DOCUMENT Not stated in the document.

NAIVE RAG ✓ CORRECT “The context does not contain the answer. The provided information only details admission requirements for the B.Des programme.”	KAG • GRAPH – NOT RUN Paraphrase variant — RAG baseline only.
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c8 Which programmes in this document require no entrance examination at all?

IN THE DOCUMENT Only B.Tech Fashion Technology. M.Des is unknown, not “none”.

NAIVE RAG ✓ CORRECT “B.Tech Fashion Technology Programme.”	KAG • GRAPH ✓ CORRECT Fashion Technology → CONFIRMED — the document explicitly states no entrance exam. M.Des → UNKNOWN — the document does not state the admission route. WHY Both look identical as missing data. Only the graph separates a stated negative from an unstated one — the open-world assumption made explicit.
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c12 Which course basket has no abbreviation defined in the document?

IN THE DOCUMENT Project and Internship.

NAIVE RAG ✓ CORRECT “Project and Internship”	KAG • GRAPH ✓ CORRECT [Project and Internship] — found by
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WHERE NOT (:Abbreviation)-
[:ABBREVIATES]->(b)

c17 Is a V-DAT score accepted for admission to M.Des?

IN THE DOCUMENT The document does not say.

NAIVE RAG ✓ CORRECT

“The context does not contain the answer.”

KAG • GRAPH ✓ CORRECT

supporting_edges: 0 , status:
NOT_STATED_IN_SOURCE

WHERE THEY DIVERGE – THE THREE MOST INTERESTING QUESTIONS ON THE PAGE

c4 On what date was FFCS Academic Regulations 4.0 approved?

IN THE DOCUMENT The date is blank in the source — it prints xxxxxxxxxxxxxxxx . Approved at the 72nd Academic Council meeting.

NAIVE RAG ✗ NUANCED

“The context does not contain the answer.”

WHY Defensible but incomplete. The retrieved chunk *does* say v4.0 was approved at the 72nd Academic Council meeting, and the model dropped it.

KAG • GRAPH ✓ CORRECT

meeting: 72nd Academic Council meeting ,
printed_date: "xxxxxxxxxxxxxxxxxx" ,
date_known: false

WHY Separates “no date was printed” from “I don’t know the date” — two different facts that prose collapses into one silence.

c5 Which entrance exam ranks are used to admit students to M.Tech?

IN THE DOCUMENT VITMEE. The source paragraph contains a typo and says “VITEEE”.

NAIVE RAG ✗ WRONG

“VITEEE ranks are used to admit students to M.Tech.”

WHY Propagated a typo in the source. The retrieved chunk literally says students are admitted “based on their VITEEE ranks” in a paragraph otherwise

KAG • GRAPH ✓ CORRECT

VITMEE (VIT Master’s Entrance Exam) ,
with a source_note recording the source’s inconsistent wording.

WHY This is the clearest case for a graph: it is where human curation lives, and the correction is

c16 How many course baskets are students allowed to choose from?

IN THE DOCUMENT Genuinely ambiguous — the source writes “Discipline Elective or Specialization Elective”.

NAIVE RAG ❌ NUANCED	KAG • GRAPH ❌ NUANCED
“Students have the option of choosing courses from 8 course baskets.”	n: 9 — the two electives modelled as separate baskets.
	WHY Not an error on either side. Building the graph forced a decision the prose left open. Which answer is right is a question about the document, not about retrieval.

GLOBAL AGGREGATES — NAIVE RAG FAILS FIVE OF SIX

c6 How many FFCS regulation versions came before version 4.0? List them in order.

IN THE DOCUMENT 7 — 1.00, 1.10, 2.00, 2.10, 3.0, 3.1, 3.2.

NAIVE RAG ✓ CORRECT	KAG • GRAPH ✓ CORRECT
“There were 7 FFCS regulation versions before version 4.0: 1.00, 1.10, 2.00, 2.10, 3.0, 3.1, 3.2”	versions_before: 7 and the same ordered list, traversed with SUPERSEDES* .
WHY The one aggregate naive RAG gets right — because all seven versions happen to sit inside a single chunk.	

c7 List all the facilities at the school that offers B.Des.

IN THE DOCUMENT PROTICS Studio, 3D-iD Studio, Smart PD Lab, Ergonomics Lab, Painting Booth.

NAIVE RAG ❌ WRONG	KAG • GRAPH ✓ CORRECT
“...facilities for design research, consultancy,	VIT School of Design, n: 5 → PROTICS

training, growth for faculty members, state-of-the-art studio/lab facilities for teaching / training / model-making / prototyping / experimenting.”

WHY Evidence split across two chunks.

chunk_p3_04 says V-SIGN offers B.Des; the five named facilities are in chunk_p3_05, which was never retrieved. The model answered from the generic prose it did have and produced a fluent list containing *none* of the real facilities.

Studio, 3D-iD Studio, Smart PD Lab, Ergonomics Lab, Painting Booth.

c9 How many Academic Council meetings are referenced in the document?

IN THE DOCUMENT 10.

NAIVE RAG ✗ WRONG

“9”

WHY Counting needs every chunk that names a meeting. They are spread over p2_03, p2_04, p2_05 and p2_06; top-4 retrieval returned p2_03, p2_04, p2_05, p2_07. One meeting was never seen, so the count came out one short.

KAG • GRAPH ✓ CORRECT

n: 10 with the full list — a `count()` over every `:CouncilMeeting` node, not over a sample.

c13 How many distinct entrance examinations are named in the document? List them.

IN THE DOCUMENT 4 — VITEEE, VITMEE, UCEED, V-DAT.

NAIVE RAG ✗ WRONG

“There are 2 distinct entrance examinations named in the document: VITEEE, VITMEE.”

WHY The four exams are introduced in four different places. Retrieval returned chunks about abbreviations and features instead of chunk_p3_06, where UCEED and V-DAT are defined — so the model saw two exams and confidently reported two.

KAG • GRAPH ✓ CORRECT

n: 4 → VITEEE, VITMEE, UCEED, V-DAT, each with its full name.

c14 List all 11 features of FFCS described in the document.

IN THE DOCUMENT Features 0 through 10.

NAIVE RAG ✗ **WRONG**

Produced eleven items by splitting single features into several and inventing others.

WHY The 11 features span four chunks; only two were retrieved. Because the question *asked for 11*, the model manufactured 11 — the instruction to produce a count overrode the evidence. This is the most instructive failure on the page.

KAG • **GRAPH** ✓ **CORRECT**

n: 11 , features 0–10 in order, sorted by f.number .

c15 Which schools are named in the document?

IN THE DOCUMENT V-SIGN (VIT School of Design) and VIT Business School.

NAIVE RAG ✗ **WRONG**

“The context does not contain the answer.”

WHY A **false negative**. “School” is topically close to the education vocabulary saturating page 1, so retrieval returned page-1 chunks about abbreviations and features. Neither school is named there, so the model correctly reported that its context lacked the answer — while the document names two.

KAG • **GRAPH** ✓ **CORRECT**

n: 2 → VIT School of Design (V-SIGN), VIT Business School.

SET 2 • GAP QUESTIONS

Where the graph fails

The mirror image. These ask about prose the ontology never modelled — rationale, objectives, description. `src/kg_coverage.py` proves the absence honestly by searching the graph and the text for the same terms and reporting *in graph? NO — in text? YES*, rather than asserting the gap. K8 – K10 (what VTOP is used for, what helps slow learners, what the document says about research for UG students) are also absent from the graph and were not put to the LLM.

K1 What do employers expect from students?

IN THE DOCUMENT Multi-disciplinary competency, leadership skills, ICT ready.

KAG · GRAPH Ø NO DATA Nothing in the graph. No entity, no edge — employer expectations were never modelled.	NAIVE RAG ✓ CORRECT “Employers expect students to have multi-disciplinary competency, leadership skills, and to be Information and Communication Technology (ICT) ready” [chunk_p1_03]
HYBRID ✓ CORRECT “Employers expect students to have multi-disciplinary competency, leadership skills, and be Information and Communication Technology (ICT) ready” [chunk_p1_03] 24 graph facts · 1 census row · 4 passages	

K2 What is the main objective of the design programmes at V-SIGN?

IN THE DOCUMENT Create a new breed of problem solvers in industrial design.

KAG · GRAPH Ø NO DATA Nothing in the graph.	NAIVE RAG ✓ CORRECT “The main objective of the programmes is to create a new breed of problem solvers in the domain of industrial design” [chunk_p3_04]
HYBRID ✓ CORRECT “...to create a new breed of problem solvers in the domain of industrial design” [chunk_p3_04] . Additionally, the programmes aim to develop students’ skills, knowledge, and aptitude to foster innovation in the product manufacturing industry [chunk_p3_05] 57 graph facts · 5 census rows · 4 passages	

K3 How are V-SIGN students trained to approach product design?

IN THE DOCUMENT Holistic viewpoint: aesthetics, ergonomics and engineering.

KAG · GRAPH

Ø NO DATA

Nothing in the graph.

NAIVE RAG

✓ CORRECT

“...from a holistic viewpoint by integrating aesthetics, ergonomics, and engineering in a balanced and harmonious manner”
[chunk_p3_05]

HYBRID

✓ CORRECT

Same quotation, **plus** the state-of-the-art studio and lab facilities used for teaching, model-making, prototyping and experimenting [chunk_p3_04]

53 graph facts · 5 census rows · 4 passages

K4 What kind of building houses the design facilities at V-SIGN?

IN THE DOCUMENT A sprawling new building.

KAG · GRAPH

Ø NO DATA

Nothing in the graph. The *facilities* are modelled; the building that holds them is not.

NAIVE RAG

✓ CORRECT

“The design studio and lab facilities at V-SIGN are housed in a sprawling new building”
[chunk_p3_04]

HYBRID

✓ CORRECT

“...housed in a ‘sprawling new building’”
[chunk_p3_04] , **plus** what the building is used for.

53 graph facts · 5 census rows · 4 passages

K5 Why does the document say present-day students need a flexible system?

IN THE DOCUMENT They like to decide and plan their future themselves.

KAG · GRAPH

Ø NO DATA

Nothing in the graph. Rationale is exactly the kind of prose an ontology projects away.

NAIVE RAG

✓ CORRECT

“...present-day students like to make decisions on their own and plan their future by themselves” [chunk_p1_03]

HYBRID

✓ CORRECT

Same, **plus** the link to employer expectations
— “multi-disciplinary competency”,
“leadership skills” — which the system serves
by letting students “build their own
curriculum” [chunk_p1_03] , [chunk_p2_01]

39 graph facts · 3 census rows · 4 passages

k6 Does being called for counselling guarantee admission to Fashion Technology?

IN THE DOCUMENT No — counselling does not guarantee admission.

KAG · GRAPH · NUANCED

Partially present. Held as a property on the
REQUIRES edge: “virtual or physical; does not
guarantee admission”.

WHY The one gap question the graph can partly
answer — because the caveat happened to be
captured as edge metadata rather than dropped.

NAIVE RAG – NOT RUN

Not part of the recorded run.

HYBRID – NOT RUN

Not part of the recorded run.

k7 In a dispute over interpreting the rules, whose interpretation is final and binding?

IN THE DOCUMENT Academic Council or Academic Policy Committee.

KAG · GRAPH · NO DATA

Nothing in the graph. The bodies exist as
nodes; their authority to interpret does not.

NAIVE RAG ✓ CORRECT

“...only the interpretation given by the
Academic Council (or) the Academic Policy
Committee will be considered final and
binding” [chunk_p2_07]

HYBRID ✓ CORRECT

Same, **plus** “the Academic Policy Committee
is headed by the Vice-Chancellor of the
University” [chunk_p2_07]

42 graph facts · 4 census rows · 4 passages

SET 3 · THE REPAIR

Hybrid on the four aggregates

The four questions naive RAG got wrong, re-run with graph facts and passages in the same prompt. The counts in each hybrid lane show how much context that took.

c13

How many distinct entrance examinations are named in the document? List them.

IN THE DOCUMENT 4 — VITEEE, VITMEE, UCEED, V-DAT.

NAIVE RAG ❌ WRONG “There are 2 distinct entrance examinations named in the document: VITEEE, VITMEE.”	HYBRID ✔️ CORRECT “There are 4 distinct entrance examinations named in the document: UCEED, V-DAT, VITEEE, and VITMEE. Dates for exams like VITEEE and VITMEE are announced through the university website or media” [chunk_p3_02] 63 graph facts · 5 census rows · 4 passages
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c15

Which schools are named in the document?

IN THE DOCUMENT VIT School of Design (V-SIGN) and VIT Business School.